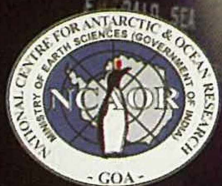


TECHNICAL PUBLICATION NO. 21



Scientific Report
TWENTY THIRD
INDIAN EXPEDITION TO
ANTARCTICA



National Centre for Antarctic & Ocean Research
(Ministry of Earth Sciences)

Goa

SCIENTIFIC REPORT OF TWENTY THIRD INDIAN ANTARCTIC EXPEDITION

TECHNICAL PUBLICATION No. 21



सत्यमेव जयते

**NATIONAL CENTRE FOR ANTARCTIC & OCEAN RESEARCH
(MINISTRY OF EARTH SCIENCES)
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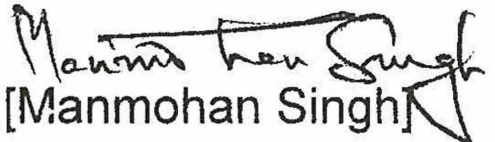
प्रधान मंत्री
Prime Minister

MESSAGE

[For Shri Srinivasan Jayaram, Station Commander and all the Members of the XXI Winter Team, 'Maitri', Antarctica]

I felicitate you and all the team members at the Indian Station 'Maitri' on the occasion of Antarctic Mid Winter Day. The National Antarctic Programme has been achieving continuous success on account of the diligent efforts made by you and your colleagues. I have no doubt that your sincere and fearless work in such harsh conditions will achieve fruition. The people of India admire the courage and determination shown by all of you.

I wish all of you a happy and successful stay in Antarctica and a safe return home.


[Manmohan Singh]

New Delhi
June 11, 2004

कपिल सिबल
KAPIL SIBAL



राज्य मंत्री (स्वतंत्र प्रभार)
विज्ञान एवं प्रौद्योगिकी तथा महासागर विकास
भारत सरकार, नई दिल्ली
MINISTER OF STATE (INDEPENDENT CHARGE) FOR
SCIENCE & TECHNOLOGY AND OCEAN DEVELOPMENT
GOVERNMENT OF INDIA,
NEW DELHI

19 April 2005

Dear Shri Jayaram,

I hope this communication finds you and other team members in the best of health and spirits.

2. The memorable moments which myself and other members of the delegation spent at Antarctica would remain vivid in our memory for long. I compliment you and your team members for the excellent arrangements made at the Indian base at Maitri and onboard M.V. Emerald Sea.

3. The National Antarctic Programme owes its success to the devotion and courage displayed by you all. Your relentless efforts in pursuit of scientific knowledge in the harsh conditions of Antarctica will certainly keep India's image bright and high in the international arena. The people of India admire your determination and grit.

4. In this connection, I would also like to place on record the appreciation of the flying skills and behaviour of Capt. Craig Rose, Senior Pilot and Mr. Bruce Towers, Pilot, the two helicopter pilots. It was amazing to see them flying the choppers in the hills and the hostile environment with such perfection.

5. Wishing you and your team members the best of luck and success in all your future endeavours.

With warm regards,

Yours sincerely,

(KAPIL SIBAL)

Shri Jayaram
23rd Expedition Leader
Indian Antarctic Expedition
C/o Director, Antarctic Division,
Geological Survey of India,
NH-VP, NIT, Faridabad



डॉ. शैलेश नायक
DR. SHAILESH NAYAK

सचिव
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FOREWORD

The icy continent of Antarctica, known for its inaccessibility, harsh climate and remoteness is a challenging place for accomplishing the scientific goals that the successive teams venturing into the unknown, set for themselves. Undoubtedly, Antarctica is the ideal laboratory to decipher and link the past with the present.

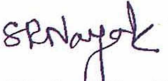
India has maintained its perceptible presence in Antarctica since 1981. Ever since, the National Centre for Antarctic & Ocean Research at Goa was entrusted with the task and declared as the nodal agency for launching these expeditions, the Antarctic Program has grown and gained from the strengths of its partners. Today most of the reputed laboratories of CSIR, DST and DRDO, Survey Organizations, IITs and leading Universities are involved with this national endeavour in some way or the other.

Each member, especially one who volunteers for wintering over, demonstrates immense grit and determination. I am happy that the XXIII Expedition has come out with success under the leadership Shri. S. Jayaram, a professional geologist from Geological Survey of India. The present report is the outcome of the results of the scientific programs undertaken by the participating organizations through their scientists under his able leadership.

This Expedition will also be remembered for the fulfilment of the stupendous assignment given to the special Task Force under the leadership of Rasik Ravindra of Geological Survey of India, to identify the site for establishing a new Research Station of India. The site has since been cleared by ATCM and the work on construction of the station has already started.

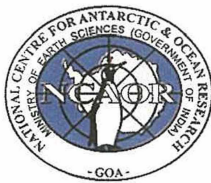
My recent visit to Antarctica in December 2009 has made me appreciate personally, the difficult conditions in which scientists and logistic personnel perform.

I congratulate the team for their achievements.


(Shailesh Nayak)

राष्ट्रीय अंटार्कटिक एवं समुद्री अनुसंधान केन्द्र

पृथ्वी विज्ञान मंत्रालय
(भारत सरकार)
हेड लैण्ड सडा, वास्को-डा-गामा
गोवा- ४०३८०४, भारत



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PREFACE

The Indian Antarctic Program has been successful in bringing into its fold, newer organizations and younger participants to revitalize this national endeavour. The *XXIII* Indian Scientific Expedition to Antarctica that sailed to the icy continent under the leadership of Shri. S. Jayaram of the Geological Survey of India had several "firsts" to its credit.

The Absolute Gravity measurements for deformation studies were undertaken by NGRI, while IIG & GSI and Survey of India carried out high precision GPS measurements for deciphering movement of the continental ice sheet and its northern front, for the first time. The GSI scientists also established an advanced camp at Gruber Mountain to map various variants of the anorthosites. It was also the year when Institute of Ayurveda participated to study the effects of ayurvedic food and health supplements on expedition members, especially as regards aging in cold climatic conditions.

The team, comprising twenty five members each for the summer and winter components, had experts from Engineers India Limited (EIL) and Structural Engineering Research Centre of CSIR (SERC) to carry out assessment of the structural strength of Station building and suggest the improvements in effluent management at Maitri.

This was also the year when a special task Force was sent to undertake a reconnaissance of coastal Antarctica to select possible site for an additional Indian research station in Eastern Antarctica.

The team and its Leader deserve appreciation for completing all the assigned tasks successfully within limited time and bringing out the current publication for the benefit of future aspirants to Antarctic Expeditions.

I would like to thank Shri S. Jayaram and all the reviewers for going through the initial draft of the report and suggesting improvements.


(Rasik Ravindra)

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THE TEAM

XXIII WINTER TEAM

Sr. No.	Name	Institution
1.	Jayaram, S.	Expedition Leader
2.	Jayapaul, D.	GSI
3.	Akilan, A.	NGRI
4.	Dilip D. Deshpande	IMD
5.	Tripathi, R.N	IMD
6.	Pavan S. Kulkarni	NPL
7.	Selvaraj, C.	IIG
8.	Manerikar, J.R.	Major (EME)
9.	Anand Kumar	Major (Engs.)
10.	Soreng, E.	EME
11.	Tejram	EME
12.	Awasthi, P.K.	EME
13.	Suresh, M.K.	EME
14.	Yadav, R.K.S.	EME
15.	Virender Singh	EME
16.	Sushil Kumar	EME
17.	Muthuraj, L.	Engineers
18.	Manoj Kumar	Engineers
19.	Sunil Kumar Singh	Engineers
20.	Ram Magan	Engineers
21.	Balbir Singh Rawat	DEAL
22.	Routela, D.P.S.	DEAL
23.	Ramesh Singh	ITBP
24.	Dr. U.S. Verma	Doctor, RLTRI
25.	Dr. V.V. Rao	Doctor, ITBP

XXIII INDIAN ANTARCTIC EXPEDITION WINTER TEAM



Srinivasan Jayaram
GSI

Leader & Station Commander



Jayapaul, D.
GSI



Akilan, A.
NGRI



D. D. Deshpande
IMD



R. N. Tripathi
IMD



Pavan S. Kulkarni
NPL



Selvaraj, C.
IIG



Manerikar, J.R.
EME



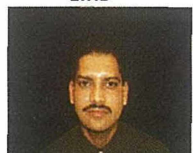
Anand Kumar
Engineers



Soreng, E.
EME



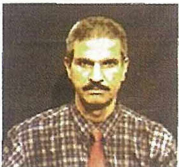
Tejram
EME



Awasthi, P.K.
EME



Suresh, P.K.
EME



Yadav, R.K.S.
EME



Virender Singh
EME



Sushil Kumar
EME

THE TEAM



Muthuraj, L.
Engineers



Manoj Kumar
Engineers



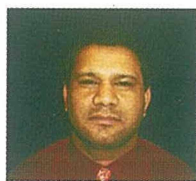
Sunil Kumar Singh
Engineers



Ram Magan
Engineers



Balbir Singh Rawat
DEAL



Dinesh P.S. Routela
DEAL



Ramesh Singh
ITBP



Dr. V. V. Rao
Doctor, ITBP



Dr. U. S. Verma
Doctor, RLTRI

THE TEAM

XXIII SUMMER TEAM

Sr. No.	Name	Institution
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2.	Golani, P.R.	GSI
3.	Amit Dharwadkar	GSI
4.	Ravikant, V.	GSI
5.	Ajay Dhar	IIG
6.	Ponraj, M.	IIG
7.	Sachin Ghude	NPL
8.	Tiwari, V.M.	NGRI
9.	Vyaghreswara Rao, M.B.S.	NGRI
10.	Abhishek Bharadwaj	DIPAS
11.	Rajesh Sanad	CCRAS
12.	Bansal, Praveen Kumar	CCRAS
13.	Palanisamy, M.	BSI
14.	Mahendra Singh	SOI
15.	Sharma, L.M.	SOI
16.	Rajan, S.	NCAOR
17.	Singh, S.M.	NCAOR
18.	Tiwari, Anoop Kumar	NCAOR
19.	Mangal Kakkad	NHO
20.	Srivastava, P.K.	NHO
21.	Rama Krishna Kopalli, S.	NHO
22.	Biswanath Gangopadhyay	DEAL
23.	Veerabhadraiah, G.	EIL

THE TEAM

Sr. No.	Name	Institution
24.	Anurag Sinha	EIL
25.	Raghava, G.	SERC
26.	Murthy, S.G.N.	SERC
27.	Ramesh Yadav	Indian Army
28.	Jagdish Kumar	Indian Army
29.	Hukum Singh Chiral	ITBP
30.	Olech, Maria	Poland

XXIII INDIAN ANTARCTIC EXPEDITION

SUMMER TEAM 2003-2004



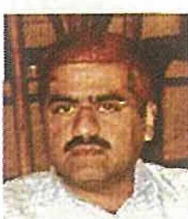
Sh Srinivasan Jayaram
GSI, Bangalore
Leader & Station Commander



Sh M B S V Rao
NGRI, Hyderabad
Summer Camp Commander



Sh Rasik Ravindra
GSI, Faridabad



Dr P R Golani
GSI, Jaipur



Sh Amit Dharwadkar
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Dr V Ravikant
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Sh Ajay Dhar
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Sh M Ponraj
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Sh A Bharadwaj
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Dr Rajesh Sannd
CRIA, Patiala



Dr P K Bansal
CRIA, Patiala



Dr M Palanisamy
BSI, Port Blair



Sh Mahendra Singh
SOI, Dehradun



Sh L M Sharma
SOI, Dehradun



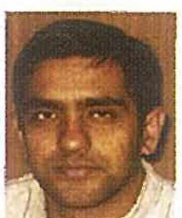
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Indian Navy



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SERC, Chennai



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Poland

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